

CASE 14

HEMOPHAGOCYTIC LYMPHOHISTIOCYTOSIS

The killing machinery of NK cells and cytotoxic T cells.

Virus-infected cells are eliminated by natural killer (NK) cells and cytotoxic CD8⁺ T cells. NK cells are lymphocytes of innate immunity that are activated to kill virus-infected cells by recognizing alterations in the MHC molecules on their surface. NK cells carry numerous activating and inhibitory receptors that between them control the cells' cytotoxicity. In contrast, the T-cell receptors on cytotoxic CD8⁺ T lymphocytes specifically recognize viral antigens presented on the infected cell's surface by MHC class I molecules. Antigen binding by the T-cell receptor activates the cytotoxic T cell. Despite the differences in recognition mechanisms, both these cell types kill their targets in the same way—by release of the cytotoxic proteins **perforin**, **granzyme B**, and **granulysin** onto the target cell (see Fig. 12.1). **Perforin** forms multimeric pores in the cell membranes that enable the delivery of the other cytotoxic proteins from the killer cell into the target.

Cytotoxic proteins are preformed and stored in endosomal 'lytic granules' within the lymphocyte. Once activated, cytotoxic lymphocytes reorient the microtubule-organizing center of the cell toward the point of contact with the target, and this cytoskeletal rearrangement guides the lytic granules to the contact point. After the lytic granules dock at the cell membrane, perforin polymerizes and inserts into the

TOPICS BEARING
ON THIS CASE:

Lymphocyte
cytotoxic activity

Macrophage
activation

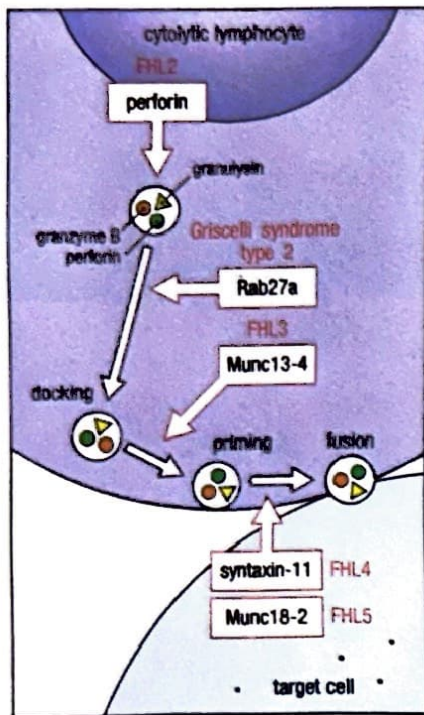


Fig. 14.1 Intracellular trafficking of lytic granules and genetic defects that can lead to hemophagocytic lymphohistiocytosis (HLH). On activation, cytotoxic CD8⁺ T cells and NK cells mobilize lytic granules filled with cytotoxic proteins that are transported toward the point of contact between the lymphocyte and the target cell, where they dock with the membrane. This is followed by the exocytosis of granule contents and the release of cytotoxic proteins through pores formed by perforin. Different steps of this process are impaired in the forms of familial hemophagocytic lymphohistiocytosis (FHL) for which the genetic defect is known (FHL 2, 3, 4, and 5) and in another disease in which HLH is a symptom (Griscelli syndrome type 2); the protein deficient in each disease is shown next to each form of FHL. A deficiency in perforin does not affect the formation of the cytolytic granules but impairs the release of the cytotoxins and their entry into the target cell.

membrane of the target cell, thus forming pores that connect the killer and target cells. Granzyme B and granulysin are released through the pore into the target cell, where they induce apoptosis (Fig. 14.1).

Several other proteins apart from the cytotoxins are crucial to successful cell-mediated cytotoxicity. The small GTPase Rab27a promotes the docking of mature cytotoxic granules to the cell membrane. The Munc13-4 protein promotes priming of the cytolytic granules; finally, syntaxin-11 and Munc18-2 (both are part of the docking complex) enable fusion of the secretory granules with the cell membrane (see Fig. 14.1). Cytotoxic proteins may then be released by exocytosis through the pores formed by perforin.

The process of degranulation can be detected by the appearance of lysosomal-membrane-associated glycoproteins on the cell membrane. In particular, surface expression of one such protein (CD107a) is often used to detect release of lytic granules (degranulation) by activated cytotoxic T cells and NK lymphocytes. Under resting conditions, CD107a is located on the inner surface of the lytic granule membrane; upon degranulation, it becomes exposed on the lymphocyte surface.

Several genetic defects that affect the cytotoxic machinery and cause disease in humans have been identified. This case concerns one of these, the disease hemophagocytic lymphohistiocytosis (HLH). Other defects cause disorders of cytotoxicity and pigmentation such as Griscelli syndrome type 2, Chediak-Higashi syndrome (see Case 15), and Hermansky-Pudlak syndrome type 2 (Fig. 14.2). All of these disorders are characterized by increased susceptibility to viral disease and an overwhelming inflammatory response, with increased production of the cytokine interferon- γ (IFN- γ).

	FHL1	FHL2	FHL3	FHL4	FHL5	Griscelli syndrome type 2	Chediak-Higashi syndrome	Hermansky-Pudlak syndrome type 2
Gene affected	Unknown	<i>PRF1</i>	<i>UNC13D</i>	<i>STX11</i>	<i>STXBP2</i>	<i>RAB27A</i>	<i>CHS1</i>	<i>AP3BP1</i>
Protein	Unknown	Perforin	Munc13-4	Syntaxin-11	Munc18-2	Rab27a	LYST	AP3B1
Function	Unknown	Pore formation	Priming	Granule fusion with cell membrane	Granule fusion with cell membrane	Docking of granules	Protein sorting	Protein sorting
HLH	+	+	+	+	+	+	+	+
Hypopigmentation	-	-	-	-	-	+	+	+
Other features	-	-	-	-	-	-	Giant lysosomes, peripheral neuropathy	Neutropenia, tendency to bleeding

Fig. 14.2 Characteristics of FHL and other diseases causing HLH.

The case of Jude Fawley: a febrile illness with altered mental status.

Jude was born after an uneventful pregnancy; his parents were distantly related. At 2 months of age he developed rhinorrhea ('running nose') and fever. Seven days later he was referred to the emergency room with a high fever (39.5°C) and difficulty in feeding. His physical examination showed significant enlargement of the liver and the spleen, which were both palpable 4 cm below the costal edge.

A complete blood count showed marked lymphocytosis ($29,000 \mu\text{l}^{-1}$), thrombocytopenia (platelet count $73,000 \mu\text{l}^{-1}$; normal $150,000\text{--}300,000 \mu\text{l}^{-1}$), and anemia (hemoglobin (Hb) 6.2 g dl^{-1} ; normal $9.0\text{--}14.0 \text{ g dl}^{-1}$). Spinal fluid sampled by lumbar puncture showed mild pleocytosis (increased white blood cell count). Laboratory investigations revealed increased levels of liver enzymes (alanine aminotransferase (ALT) 375 U l^{-1} ; aspartate aminotransferase (AST) 220 U l^{-1}), markedly high ferritin (7500 ng dl^{-1} ; normal $50\text{--}200 \text{ ng dl}^{-1}$), positive C-reactive protein, and elevated fibrin degradation products.

Serological tests for Epstein-Barr virus, cytomegalovirus, and adenovirus were all negative. Although blood cultures were negative, Jude was treated with intravenous antibiotics (ampicillin and amikacin), but his condition continued to worsen, and his consciousness rapidly deteriorated. Brain magnetic resonance imaging (MRI) with T2 intensity revealed spotty high-density lesions in the white matter of the cerebrum, basal nuclei, and cerebellum. Laboratory investigations showed a worsening of anemia (Hb 5.2 g dl^{-1}) and thrombocytopenia ($31,000 \mu\text{l}^{-1}$), but also high serum levels of triglycerides (750 mg dl^{-1} ; normal value less than 150 mg dl^{-1}) and low fibrinogen (40 mg dl^{-1} ; normal value $200\text{--}400 \text{ mg dl}^{-1}$). Bone marrow aspiration was performed and showed hypocellularity with an increased number of large granular lymphocytes and macrophages with hemophagocytic activity. Jude was diagnosed with HLH. Flow cytometry analysis revealed an absence of perforin. Complete absence of NK cell-mediated cytotoxicity was demonstrated using a suitable target (K562) cell line. Sequence analysis of the perforin gene revealed a single nucleotide deletion in exon 2, resulting in a frameshift and the production of an aminoterminally truncated perforin protein that could not be detected by flow cytometry analysis. Both parents were heterozygous for the same mutation.

Jude received chemotherapy according to the HLH-94 protocol, which consisted of dexamethazone, etoposide, and cyclosporin A (CsA). He subsequently underwent matched unrelated hematopoietic stem-cell transplantation (HSCT) with a reduced intensity conditioning regimen based on fludarabine, melphalan, and alemtuzumab (a monoclonal antibody directed against CD52, which is expressed by mature lymphocytes but not by stem cells), and CsA and methylprednisolone for graft-versus-host disease prophylaxis. Three years after HSCT, Jude is doing well, with mild mental and motor delay in his development.

Hemophagocytic lymphohistiocytosis.

The case of Jude Fawley illustrates the typical features of HLH. Both congenital and acquired forms of HLH are known; Jude had the congenital form, which is often known as FHL. Acquired HLH may be secondary to infections, malignancies, or autoimmune diseases. In particular, patients with juvenile arthritis or systemic lupus erythematosus can develop a disease similar to HLH, called macrophage activation syndrome. HLH is an aggressive and potentially life-threatening disease.

FHL is a rare disorder, with an estimated frequency of 1 in 50,000 births in the United States. The disease typically appears in infancy or early childhood; both

Fever and
hepatosplenomegaly.

Coagulopathy and markedly
increased ferritin.

Hemophagocytosis in the
bone marrow; no NK-cell
cytotoxicity.

sexes are affected, which is consistent with autosomal recessive inheritance. The clinical presentation is characterized by high and persistent fever, and spleen and liver enlargement. Neurological manifestations (ranging from seizures and confusion to coma) are common and are associated with pleocytosis in the cerebrospinal fluid, also known as lymphocytic meningitis. Patients develop severe anemia and thrombocytopenia, abnormal liver function, and coagulopathy (decreased fibrinogen and increased levels of fibrin degradation products, which make the blood less able to clot). Increased levels of serum triglycerides and of inflammatory markers (ferritin and C-reactive protein) are also observed.

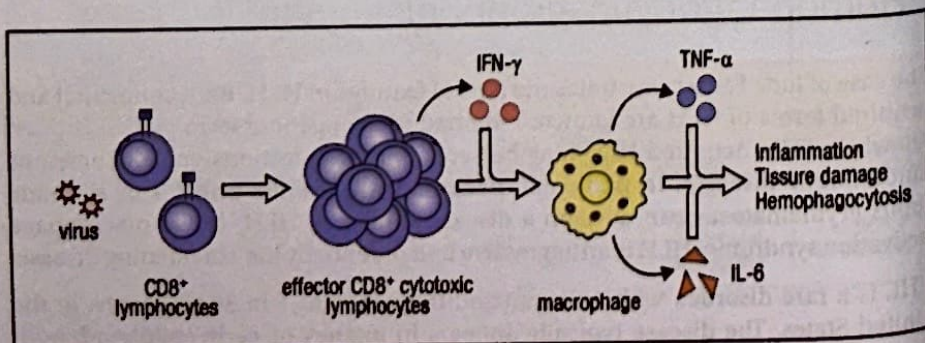
These manifestations are the consequence of the defect in lymphocyte cytotoxicity that makes patients with HLH unable to kill virus-infected cells. In this situation, there is continuous activation of NK cells and CD8⁺ T lymphocytes, which infiltrate the liver, spleen, bone marrow, and central nervous system and secrete high amounts of IFN- γ . This cytokine is a potent activator of macrophages, which are induced to secrete pro-inflammatory cytokines such as IL-6 and TNF- α . The hemophagocytosis—the phagocytic destruction of red blood cells—is also a consequence of macrophage activation (Fig. 14.3). An increase in serum levels of the soluble IL-2 receptor (sIL-2R) is also characteristic of HLH during active phases of the disease, and is a marker of T-cell activation.

In FHL, the clinical symptoms of fever, immune activation, and increased inflammation are precipitated especially by infections with herpesviruses, such as cytomegalovirus, Epstein-Barr virus, and varicella-zoster virus. These episodes are known as the 'accelerated phase' of the disease. They can lead to multiple organ failure and death, and can occur multiple times in a patient's life.

At least five variants of FHL are known (see Fig. 14.2), and the underlying genetic defect is known in four of them. Jude had FHL2, and was unable to produce a functional perforin. Some forms of HLH (Griscelli syndrome type 2, Chediak-Higashi syndrome, and Hermansky-Pudlak syndrome type 2) are also characterized by hypopigmentation, reflecting the role of the affected proteins in melanogenesis (melanin production). Patients with Chediak-Higashi syndrome also exhibit peripheral neuropathy and their leukocytes have characteristically large lysosomes (see Case 15), whereas patients with Hermansky-Pudlak syndrome type 2 show neutropenia (low levels of circulating neutrophils) and increased bleeding due to functionally abnormal platelets. Finally, hemophagocytic lymphohistiocytosis is often observed in patients with X-linked lymphoproliferative syndrome type 1 or type 2 (see Case 13), IL-2-inducible T cell kinase (ITK) deficiency (see Case 54), and CD27 deficiency.

The diagnosis of HLH is based on a combination of clinical and laboratory features. In FHL, there is a genetically determined defect in lymphocyte-mediated cytotoxicity. NK cell-mediated cytotoxicity is usually assessed by culturing the patient's peripheral blood mononuclear cells with ⁵¹Cr-labeled K562 target cells. If the patient's NK cells have intact cytotoxic activity, ⁵¹Cr is released into the supernatant. Any defect in activation, docking, or priming of the lytic granules can be detected by flow-cytometric analysis for the appearance of CD107a on the surface

Fig. 14.3 Cytokine storm in HLH. In patients with HLH, viral infections trigger the activation of CD8⁺ T cells that undergo clonal expansion and activation, but cannot clear the viral infection. The activated CD8⁺ T cells secrete large amounts of IFN- γ , which promotes the activation of macrophages and the release of TNF- α , IL-6, and other pro-inflammatory molecules. This series of events ultimately causes tissue damage and hemophagocytosis.



of the patient's lymphocytes activated *in vitro*. This test gives a normal result in patients such as Jude, whose specific deficiency is in perforin. Flow cytometry may also be used to diagnose FHL2, by means of intracellular staining for perforin.

Treatment of FHL is based on aggressive immunosuppression to stop the ongoing immune activation. However, even if this is sufficient to achieve clinical remission, patients remain highly prone to other episodes of accelerated phase, and the disease has a very high mortality rate. The only cure is through HSCT, which gives better results if reduced intensity conditioning is used. Experience in animal models suggests that administration of anti-IFN- γ monoclonal antibody might be effective to achieve remission, without the risk of side effects related to the use of chemotherapy, steroids, or other immunosuppressive drugs. Administration of anti-human IFN- γ for patients with HLH is currently being evaluated in a clinical trial. For the nonfamilial, secondary forms of HLH, treatment is based on elimination of the trigger, in addition to immune suppression.

Questions.

- 1 Why do patients with FHL develop hyperinflammatory responses in spite of their immune deficiency?
- 2 Why did Jude develop enlarged liver and spleen?
- 3 Why did Jude show hypocellularity in his bone marrow?